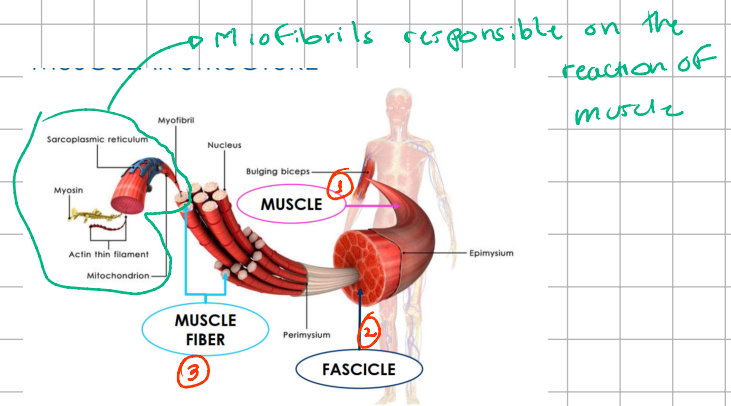
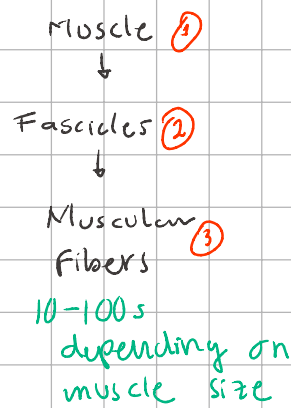


Practice 1 - Brief Recap

EMG signals → signals from the muscle.



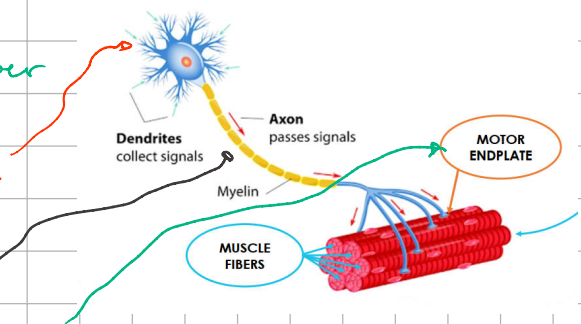
Each motor fiber is connected to a motor neuron which is the responsible for the contraction of the muscle

Motor Unit: Motor neuron + Muscle Fiber

Basically a signal is generated at the level of the spinal cord

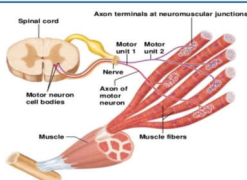
↳ it travels through the axon

↳ And then innervates the muscular fibers, through the motor endplate



One muscle is innervated by different motor neurons

Motor neuron pool: cluster of motor neurons that innervate the same muscle



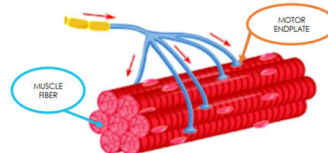
One motor neuron innervates different fibres of one muscle

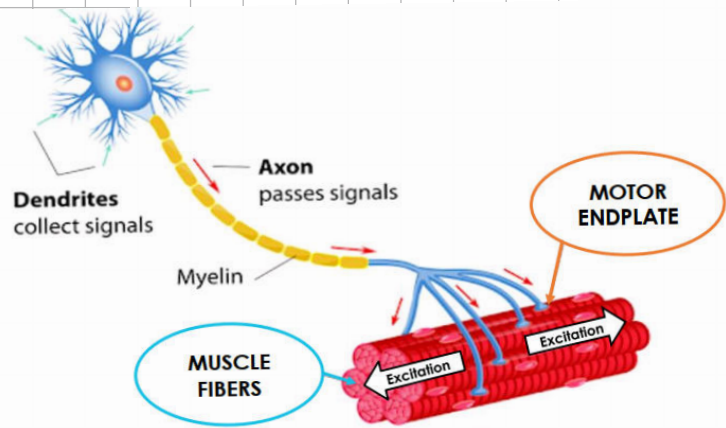
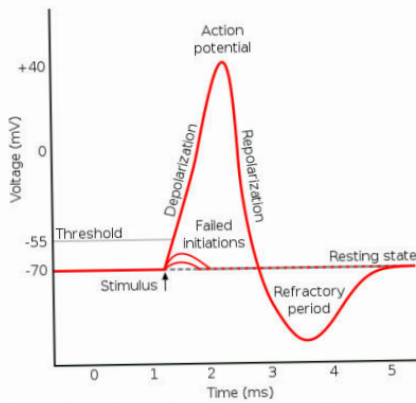
« All or none law »: the neuron spikes and all fibres innervated contract

Innervation ratio: number of fibers innervated by a single motor neuron → **Power!**

One fiber is innervated by one and only one motor neuron

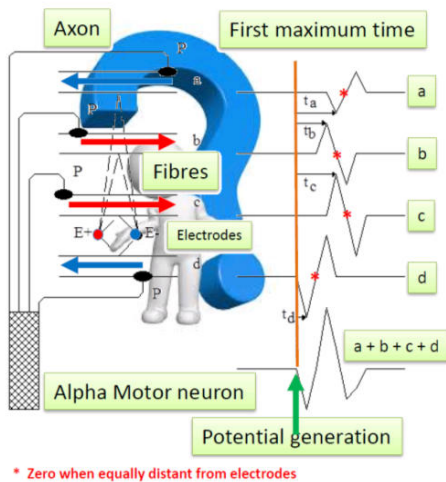
One and only one endplate per muscular fiber





Action potential takes place once signal coming from the spinal cord travels through motor neuron axon to innervate muscle fibers and trigger AP, however, not only one action potential but a many in a motor unit, because one motor neuron can be attached to different muscular fibers

MOTOR UNIT ACTION POTENTIAL (MUAP)



- What happens by changing the position of electrodes?
- **MUAP changes by changing electrodes' configuration.**

$$t_d = \frac{D}{v_c}$$

t_d : delay between AP generation and AP detection
 D : distance between motor endplate and detection location
 v_c : fiber conduction velocity

$$A = \frac{a}{d^2} \cdot A_0$$

A : sensed signal amplitude
 a : propagation coefficient
 d : transversal distance from the electrode (vertical one)
 A_0 : max signal amplitude

And we compute it (the MUAP) by means of

$$MUAP = \sum_{k=1}^K A P_k$$

K : number of fibers in the motor unit

P : Endplate

A : Sensed signal amplitude

Summation of the MUAP or MUAP * $\delta(t)$

$$MUAPT_i(t) = \sum_{k=1}^K MUAP_i(t - \tau_{k,i})$$

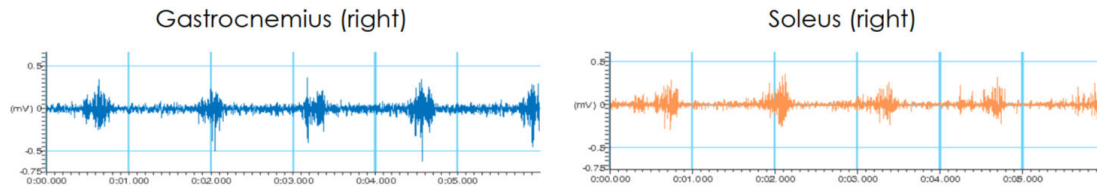
t : time instant
 $\tau_{k,i}$: time delay
 i : index of MU
 k : time index

$$MUAPT_i(t) = \sum_k MUAP_i * \delta(t - \tau_{k,i})$$

δ : Diract delta function

Motor Unit Action Potential spikes at certain freq and this is given by MUAPT which is modeled by these two

$$E(t) = \sum_i \text{MUAPT}_i(t) = \sum_i \sum_k \text{MUAP}_i(t - \tau_{k,i})$$



We obtain as a result the EMG signal!!!
 summation of MUAPT which is stochastic noisy signal!

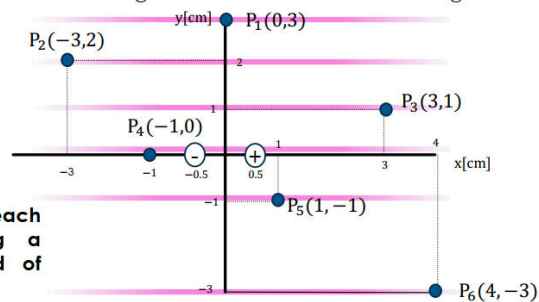
Exercises !!

EXERCISE 1

A bipolar electrode configuration is connected to an ideal differential OpAmp to detect the EMG originated by a unique motor unit. The latter contemplates six endplates: P1,...,P6, disposed as shown in figure with respect to the positive and negative electrodes, placed respectively at (0.5; 0)cm and (-0.5; 0) cm. Supposing that no signal attenuation occurs along the fibre and an activation signal with the following characteristics:

- Equal for all the fibres
- Positive with amplitude $A_0 = 10\text{mV}$
- Monophasic
- Duration: 4ms

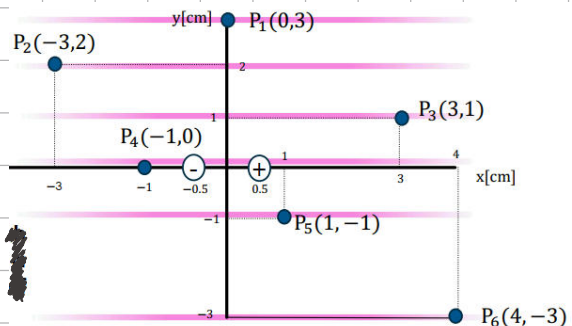
Estimate and draw the contributes coming from each endplate and the whole MUAP considering a propagation coefficient $a = 0.9$ and a speed of propagation $v_c = 0.25 \frac{\text{cm}}{\text{ms}}$.



What we need to do: Estimate and draw contributes coming from the endplates of the muscle fibers and for this we need to compute t_d and A which are the two parameters that vary depending on the location of the electrodes

$$t_d = \frac{0}{0.25 \frac{\text{cm}}{\text{ms}}}$$

Horizontal Distance between an endplate and the closest electrode



$$t_d = \left| \frac{1 \times 1 - 0.5}{0.25} \right| = |4 \times 1 - 2|$$

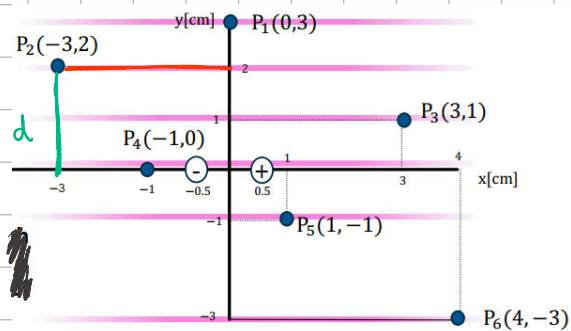
$$t_{d1} = |4 \times 1 - 2|, |4 \times 1 + 2|$$

$t_{d2} - t_{d1} = 4\text{ms}$, latency between the two electrodes

Example, P4 is gonna be firstly detected by $\ominus$ and 4ms after by $\oplus$

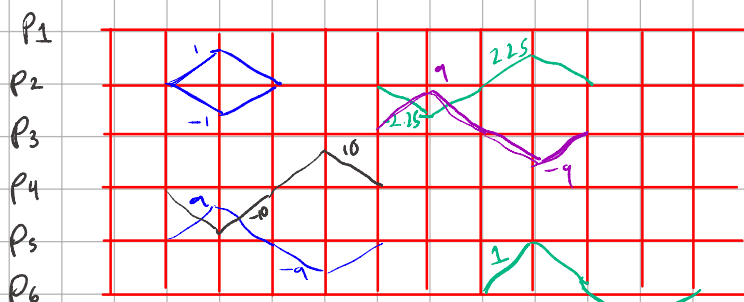
Amplitude $A_p = \frac{A_0}{d^2}$ $\rightarrow$ transversal distance

Delay $t_d = |4|x_1| - 2|$

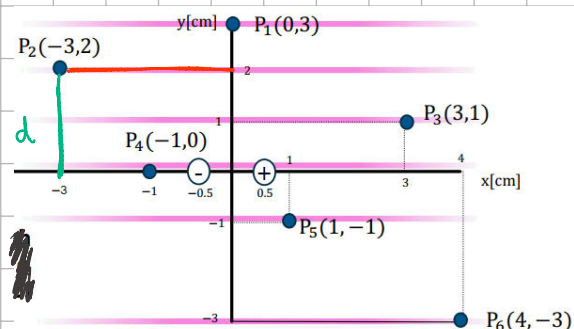
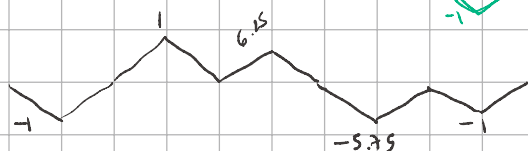


End Place	Amplitude	Delay
$P_1(0,3)$	$A_{P1} = 0.9 \times 10 / 3^2 = 1 \text{ mV}$	$\tau_1 = 4 0 - 2 = 2 \text{ ms}$
$P_2(-3,2)$	$A_{P2} = 0.9 \times 10 / 2^2 = 2.25 \text{ mV}$	$\tau_2 = 4 -3 - 2 = 10 \text{ ms}$
$P_3(3,1)$	$A_{P3} = 0.9 \times 10 / 1^2 = 9 \text{ mV}$	$\tau_3 = 4 3 - 2 = 10 \text{ ms}$
$P_4(-1,0)$	$A_{P4} = 10 \text{ mV}$	$\tau_4 = 4 -1 - 2 = 2 \text{ ms}$
$P_5(1,-1)$	$A_{P5} = 0.9 \times 10 / (-1)^2 = 9 \text{ mV}$	$\tau_5 = 4 1 - 2 = 2 \text{ ms}$
$P_6(4,-3)$	$A_{P6} = 0.9 \times 10 / (-3)^2 = 1 \text{ mV}$	$\tau_6 = 4 4 - 2 = 14 \text{ ms}$

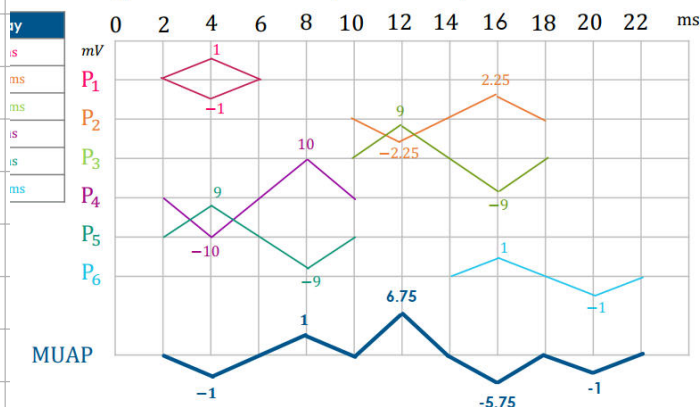
0 2 4 6 8 10 12 14 16 18 20 22 ms



MUAP



P_1 is symmetrically placed between $\ominus$ and $\oplus$ thus start both and signal lasts 4 ms



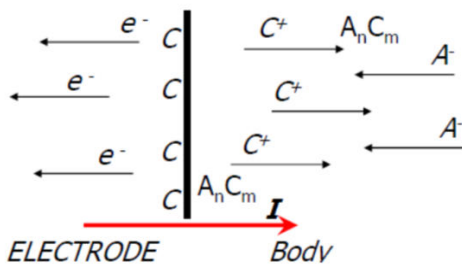
Brief Recap 2.

How EMG signal is detected? → Through Electrodes.

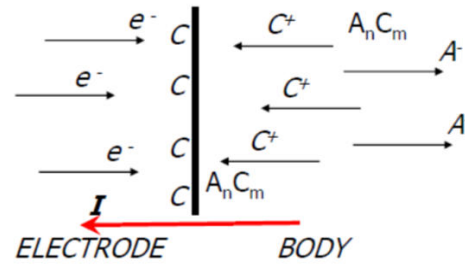
Oxidation ↔ Reaction → Provides an electrical signal

Electrodes sensors which transform ionic currents circulating within the human body in electronic currents in a metal conductor.

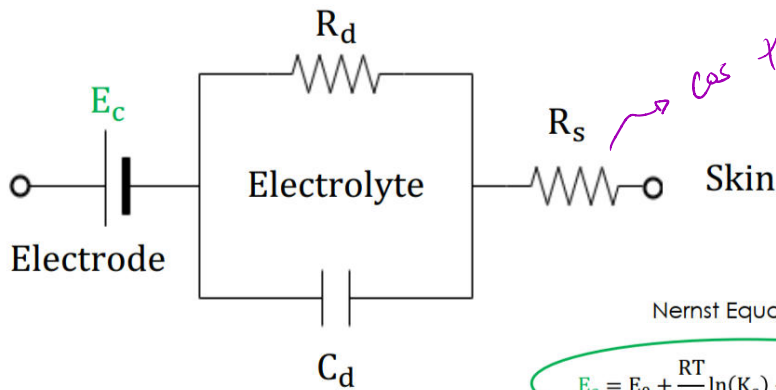
Oxidation is prevalent



Reduction is prevalent



Electrode is a metal and Electrolyte is essentially a gel which aims to reduce impedance.



Nernst Equation

$$E_c = E_0 + \frac{RT}{nF} \ln(K_s) - \frac{RT}{nF} \ln(\alpha_{Cl^-})$$

E_c : contact potential

R_d : resistance between capacitor plates

C_d : electrode-electrolyte interface stray capacitance

R_s : electrode-electrolyte resistance

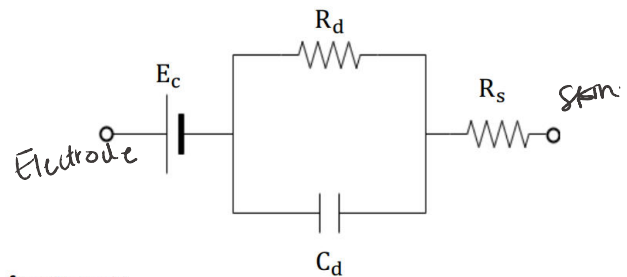
as they don't travel at light speed.

Exercises!!!

EXERCISE 2

Considering the impedance of the skin-electrode surface as described in the schematic and assuming:

- $R_d = 1 \text{ M}\Omega$
- $R_s = 120 \Omega$
- $C_d = 10 \text{ nF}$



2.1. Discuss the impedance value with respect to the frequency.

2.2. Consider the EMG bandwidth between 0-500 Hz, does this electrode work for EMG acquisition?

Starting from computing impedance w.r.t frequency,

$$Z_{total} = (Z_{Rd} \parallel Z_{Cd}) + Z_{Rs} \quad Z_{Cd} = \frac{1}{j\omega C_d}$$

$$Z_{total} = \frac{R_d \cdot \frac{1}{j\omega C_d}}{R_d + \frac{1}{j\omega C_d}} + R_s \rightarrow \frac{R_d}{R_d j\omega C_d + 1} + R_s$$

$$Z_{total} = \frac{R_d + R_s R_d j\omega C_d + R_s}{R_d j\omega C_d + 1}$$

Now when $\omega \rightarrow 0$

$$Z_{total} = R_d + R_s$$



$$Z_{total} = 1 \text{ M}\Omega + 120 \Omega \approx 1 \text{ M}\Omega$$

when $\omega \rightarrow \infty$

$$Z_{total} = R_s$$

$$Z_{total} = 120 \Omega$$

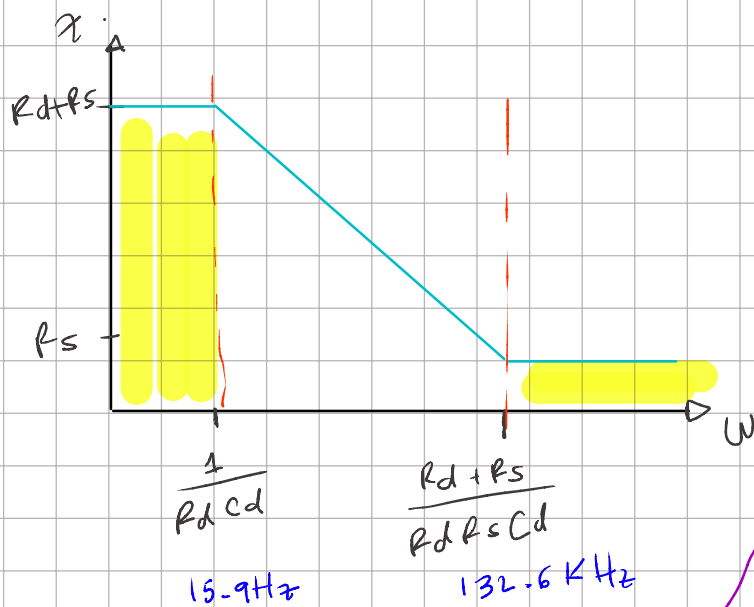


Now let's compute pole and zero

For pole $R_d C_d j\omega + 1 = 0 \rightarrow R_d C_d j\omega = -1 \quad \omega = \frac{1}{R_d C_d}$

For zero $R_d + R_s R_d j\omega C_d + R_s = 0$

$$\hookrightarrow R_s R_d C_d j\omega - (R_s + R_d) \quad \omega = \frac{R_s + R_d}{R_s R_d C_d}$$



EMG BW goes from 0-500 Hz.
Does R_d , R_s , C_d suitable to have a constant impedance.

$$\omega_p = 100 \frac{\text{rad}}{\text{s}} \rightarrow f_p = \frac{\omega_p}{2\pi}$$

$$f_p = 15.9 \text{ Hz}$$

$$\omega_z = 833430 \frac{\text{rad}}{\text{s}} \rightarrow f_z = \frac{\omega_z}{2\pi}$$

$$f_z = 132.6 \text{ KHz}$$

We aim to have a constant impedance and it happens if $f_p > 500 \text{ Hz}$ or $f_z < 10 \text{ Hz}$.

Solution

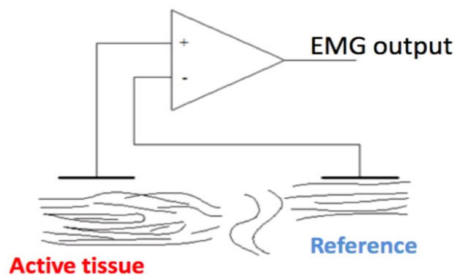
$$\rightarrow R_d = 10 \text{ K}\Omega, R_s = 120 \Omega, C_d = 10 \text{ nF}$$

$$\omega_p = 10000 \frac{\text{rad}}{\text{s}} \rightarrow f_p = 1.59 \text{ KHz} \checkmark$$

$$\omega_z = 843333 \frac{\text{rad}}{\text{s}} \rightarrow f_z = 134.22 \text{ MHz}$$

Brief Recap Electrodes Configuration

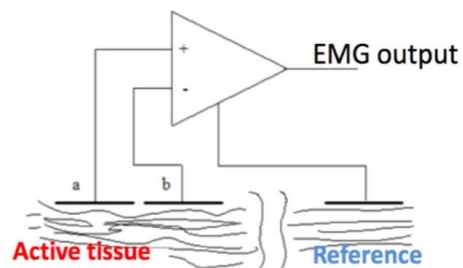
Monopolar amplifier



- ✗ Low resolution
- ✗ Noise amplification

Cannot distinguish between signal and noise

Bipolar amplifier (differential)



- ✓ High resolution
- ✓ Common mode noise rejection

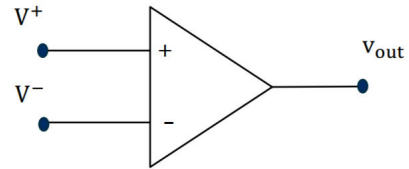
Both electrodes on tissue and high CMRR

BIPOLAR AMPLIFIER - DIFFERENTIAL

Common mode signal $V_c = \frac{V^+ + V^-}{2}$

Differential mode signal $V_d = V^+ - V^-$

$$V_{out} = V_{out,D} + V_{out,C} = A_d(V^+ - V^-) + A_c\left(\frac{V^+ + V^-}{2}\right)$$



Common mode rejection ratio $CMRR = \frac{A_d}{A_c}$

$$CMRR_{dB} = 20 \log_{10} \frac{A_d}{A_c}$$

✓ 120-140 dB

Exercise!!!

EXERCISE 3

Calculate in dB the CMRR of a differential amplifier on whose input there is a surface EMG differential signal of $1mV_{pp}$ superimposed on a common mode of $1V_{pp}$ in order to obtain on the output a ratio between $V_{out,c}$ and $V_{out,d}$ of at least 0.1. Calculate the CMRR also in the case of $V_d = 10\mu V_{pp}$.

$$V_{dpp} = 1mV \quad V_{cpp} = 1V \quad \frac{V_{out,c}}{V_{out,d}} = 0.1$$

$$CMRR_{1mV} = ?$$

$$CMRR_{10\mu V} = ?$$

$$A_d = \frac{V_{out,d}}{V_{dpp}} = \frac{V_{out,d}}{1 \times 10^{-3}}$$

$$A_c = \frac{V_{out,c}}{V_{cpp}} = \frac{V_{out,c}}{1}$$

We don't know the V_{out} in both cases, but we know their ratio.

$$CMRR_{1mV} = \frac{A_d}{A_c} = \frac{\frac{V_{out,d}}{1 \times 10^{-3}}}{\frac{V_{out,c}}{1}} = \frac{V_{out,d}}{V_{out,c} \cdot 1 \times 10^{-3}}$$

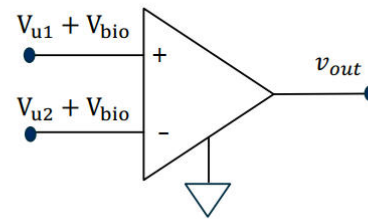
$$CMRR_{1mV} = \frac{1}{0.1} \cdot \frac{1}{1 \times 10^{-3}} = 10^4 \rightarrow CMRR_{1mV dB} = 20 \cdot \log(10^4) = 80 \text{ dB}$$

$$CMRR_{10\mu V} = \frac{1}{0.1} \cdot \frac{1}{10 \times 10^{-6}} = 10^6 \rightarrow CMRR_{10\mu V dB} = 20 \log(10^6) = 120 \text{ dB}$$

EXERCISE 4

At a generic instant t , a generic disturbance superimposes on the useful signals:

- Useful signals:
 - $V_{u1} = 3 \text{ mV}$
 - $V_{u2} = 2.9 \text{ mV}$
- Noise signal:
 - $V_{bio} = 4 \text{ mV}$



4.1. Compute the amplifier parameters (A , CMRR) in order to obtain an ideal differential output of 5 V and a ratio between the common signal and the differential signal equal to 0.04 .

4.2. The relative error introduced by the presence of the BIO signal wrt the useful one.

$$V_{out} = 5 \text{ V} \quad \text{and} \quad \frac{V_{out,c}}{V_{out,d}} = 0.04$$

$$V_{out,d} = A_d (V^+ - V^-) = A_d (V_{u1} + \cancel{V_{bio}} - V_{u2} - \cancel{V_{bio}}) \rightarrow V_{out,d} = A_d (V_{u1} - V_{u2})$$

$$V_{out,d} = A_d \cdot 0.1 \text{ mV} = 5 \text{ V}$$

$$V_{out,c} = A_c \cdot \frac{(V^+ + V^-)}{2} = \frac{A_c}{2} \cdot (V_{u1} + V_{bio} + V_{u2} + V_{bio})$$

$$V_{out,c} = A_c \cdot 6.95 \text{ mV}$$

$$A_d = 50 \times 10^3$$

$$V_{out,c} = 0.04 \times 5 = 0.2 \text{ V}$$

$$A_c = 28.7$$

Now

$$\text{CMRR}_{\text{dB}} = 20 \log_{10} \left(\frac{50 \times 10^3}{28.7} \right) = 64.8 \text{ dB}$$

Now the error introduced by V_{bio}

$$V_{out, \text{tot}} = V_{out,d} + V_{out,c}$$

with noise

$$V_{out, \text{noise}} = 5 + 0.2 = 5.2 \text{ V}$$

without noise

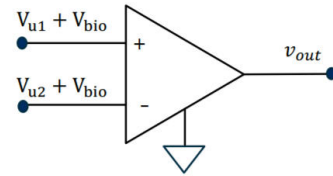
$$V_{out, \text{useful}} = A_d (V_{u1} - V_{u2}) + \frac{A_c}{2} (V_{u1} + V_{u2}) = 5.085 \text{ V}$$

$$\text{Error} \% = \frac{5.2 - 5.085}{5.085} = 0.022 = 2.2 \%$$

HOMEWORK

The two inputs of an EMG bipolar system are:

- $V_{u1} = 3.8 \text{ mV}$
- $V_{u2} = 3.3 \text{ mV}$



1.1. Determine the OpAmp parameters (A_d , CMRR) such that $V_{out,d} = 8 \text{ V}$ and the ratio between common mode output $V_{out,c}$ and differential mode output $V_{out,d}$ is equal to 2%

$A_d = ?$ CMRR = ?

1.2. Supposing acceptable a relative error of 3% between useful signal and the one obtained by summing the useful signal to a biological source $V_{useful} + V_{bio}$, estimate the maximum contribute of the biological source V_{bio} to the two components V_{u1} and V_{u2} .
 $V_{bio} = ?$

$$V_{out,d} = A_d (V^+ - V^-) = A_d (V_{u1} + \cancel{V_{bio}} - V_{u2} - \cancel{V_{bio}})$$

$$V_{out,d} = A_d (V_{u1} - V_{u2})$$

$$A_d (3.8 \text{ mV} - 3.3 \text{ mV}) = 8 \text{ V}$$

$$A_d = \frac{8 \text{ V}}{5 \times 10^{-4} \text{ V}} = 16 \times 10^3$$

$$\frac{V_{out,c}}{V_{out,d}} = 0.02$$

$$V_{out,c} = 0.02 \cdot V_{out,d} \rightarrow V_{out,c} = 0.02 \cdot 8 = 0.16$$

$$V_{out,c} = A_c \cdot \frac{(V_{u1} + V_{u2})}{2} \rightarrow A_c = \frac{0.16 \cdot 2}{(3.8 + 3.3) \times 10^{-3}} = 45.07$$

$$\text{CMRR}_{\text{dB}} = 20 \log_{10} \left(\frac{16 \times 10^3}{45.07} \right) = 51 \text{ dB}$$

$$V_{\text{noise}} = A_d (V_{u1} - V_{u2}) + \frac{A_c}{2} (V_{u1} + V_{u2} + 2V_{bio})$$

$$V_{\text{useful}} = A_d (V_{u1} - V_{u2}) + \frac{A_c}{2} (V_{u1} + V_{u2}) = 8 \text{ V} + 0.16 \text{ V}$$

$$\% \text{ Error} = \frac{V_{\text{noise}} - V_{\text{useful}}}{V_{\text{useful}}} = \frac{A_c \cdot V_{bio}}{8.16 \text{ V}} = 0.03$$

$$V_{bio} = \frac{0.03 \cdot 8.16 \text{ V}}{45.07} = 5.43 \text{ mV}$$